

SEO Analytics and Traffic Prediction Project

1. Project Overview

The objective of this project was to analyze SEO keyword performance data and develop a machine learning model to predict website traffic. The project integrates PostgreSQL, SQL, Power BI, Python, and Machine Learning to generate actionable SEO insights.

2. Dataset Description

The dataset contains keyword-level SEO metrics, including:

- Keyword
- Search Volume
- Traffic
- CPC
- Position
- Previous Position
- Keyword Difficulty
- Competition
- Traffic Cost
- Keyword Intent
- SERP Features

Total Records: 29,880

3. Tools and Technologies

- PostgreSQL
 - pgAdmin
 - SQL
 - Power BI
 - Python
 - Pandas
 - NumPy
 - Matplotlib
 - Scikit-learn
 - Google Colab
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4. Data Preparation

The dataset was imported into PostgreSQL and connected to Power BI.

Data cleaning activities included:

- Handling missing values
 - Verifying table relationships
 - Removing blank records
 - Data type validation
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5. Exploratory Data Analysis (EDA)

Several visualizations were created to understand keyword performance:

Search Volume Distribution

Observation: The distribution is highly right-skewed. Most keywords have relatively low search volume, while a small number of keywords account for extremely high search demand.

Recommendation: Focus on high-volume keywords while maintaining a balanced portfolio of medium-volume keywords.

Keyword Position Distribution

Observation: Most keywords rank within top search positions, indicating strong SEO visibility.

Recommendation: Prioritize keywords ranking between positions 4 and 10 to maximize traffic gains.

Keyword Intent Analysis

Observation: Informational keywords generate the largest share of traffic.

Recommendation: Increase content targeting commercial and transactional intent keywords to improve conversion opportunities.

Correlation Analysis

Observation: Search Volume demonstrated the strongest positive relationship with Traffic.

Recommendation: Prioritize keyword opportunities based on search demand.

6. Feature Engineering

The following features were created:

- Ranking Change
- Position Bucket
- Top 10 Ranking Flag
- Traffic Efficiency

These features were designed to improve analytical and predictive performance.

7. Feature Selection

Random Forest Feature Importance was used to identify the most influential variables.

Results indicated:

- Search Volume was the dominant predictor of Traffic.
 - CPC had minimal influence.
 - Other SEO metrics contributed comparatively little predictive value.
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8. Machine Learning Models

Three regression models were trained and evaluated:

1. Linear Regression
2. Random Forest Regressor
3. Gradient Boosting Regressor

Evaluation Results:

- Linear Regression: $R^2 = 0.9992$
- Random Forest: $R^2 = 1.0000$
- Gradient Boosting: $R^2 = 1.0000$

Random Forest achieved the lowest prediction error and was selected as the best-performing model.

9. Key Findings

- Search Volume is the strongest driver of Traffic.
- High-volume keywords provide the greatest growth opportunities.
- Random Forest produced the most accurate traffic predictions.
- Most keywords already rank within strong search positions.

- Informational intent dominates the keyword portfolio.
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10. Business Recommendations

- Prioritize high-search-volume keywords.
 - Improve rankings for keywords positioned between 4 and 10.
 - Expand content targeting commercial and transactional intents.
 - Monitor top-ranking keywords to maintain visibility.
 - Incorporate additional SEO metrics such as CTR, backlinks, and impressions for future predictive modeling.
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11. Conclusion

This project successfully combined database management, business intelligence, and machine learning techniques to analyze SEO performance and predict website traffic. The results demonstrate that keyword search demand is the primary factor influencing organic traffic and provide a framework for data-driven SEO decision-making.